

1. main.c

```
#include <stdio.h>
```

```
#include "polynomial.h"
```

```
#include <stdlib.h>
```

```
#include <string.h>
```

```
int main(){
```

```
    poly p1, p2, p3;
```

```
    int n1, n2;
```

```
    printf("Enter maximum no of terms in polynomial 1: ");
```

```
    scanf("%d", &n1);
```

```
    printf("Enter maximum no of terms in polynomial 2: ");
```

```
    scanf("%d", &n2);
```

```
    init_poly(&p1, n1);
```

```
    init_poly(&p2, n2);
```

```
    init_poly(&p3, n1 + n2);
```

```
    for(int i = 0; i < n1; i++){
```

```
        int c, e;
```

```
        printf("Enter coefficient and power of term %d of polynomial 1\n", i + 1);
```

```
        scanf("%d%d", &c, &e);
```

```
        append(&p1, c, e);
```

```
    }
```

```
    for(int i = 0; i < n2; i++){
```

```
        int c, e;
```

```
        printf("Enter coefficient and power of term %d of polynomial 2\n", i + 1);
```

```
        scanf("%d%d", &c, &e);
```

```
        append(&p2, c, e);
```

```
    }
```

```
    printf("Polynomial 1: ");
```

```
display(p1);
```

```
printf("Polynomial 2: ");
```

```
display(p2);
```

```
addition_polynomial(&p1, &p2, &p3);
```

```
printf("Sum of Polynomials: ");
```

```
display(p3);
```

```
subtraction_polynomial(&p1, &p2, &p3);
```

```
printf("Difference of Polynomials: ");
```

```
display(p3);
```

```
return 0;
```

```
}
```

2. polynomial.h

```
typedef struct term {
```

```
    int coef;
```

```
    int exp;
```

```
} term;
```

```
typedef struct poly {
```

```
    int n;
```

```
    term *t;
```

```
} poly;
```

```
void init_poly(poly *p, int max_terms);
```

```
void append(poly *p, int coef, int exp);
```

```
void display(poly p);
```

```
void addition_polynomial(poly *p1, poly *p2, poly *p3);
```

```
void subtraction_polynomial(poly *p1, poly *p2, poly *p3);
```

3. polynomial.c

```
#include <stdio.h>
```

```
#include "polynomial.h"
```

```
#include <stdlib.h>
```

```
#include <string.h>
```

```
void init_poly(poly *p, int max_terms){
```

```
    p -> n = 0;
```

```
    p -> t = (term *)malloc(max_terms * sizeof(term));
```

```
}
```

```
void append(poly *p, int coef, int exp){
```

```
    p -> t[p -> n].coef = coef;
```

```
    p -> t[p -> n].exp = exp;
```

```
    p -> n++;
```

```
}
```

```
void display(poly p){
```

```
    for(int i = 0; i < p.n; i++){
```

```
        if(p.t[i].coef < 0){
```

```
            printf(" - %dx^%d", - (p.t[i].coef), p.t[i].exp);
```

```
        }else{
```

```
            if(i > 0){
```

```
                printf(" + ");
```

```
            }
```

```
            printf("%dx^%d", p.t[i].coef, p.t[i].exp);
```

```
        }
```

```
    }
```

```
    printf("\n");
```

```
}
```

```

void addition_polynomial(poly *p1, poly *p2, poly *p3){
    int i = 0, j = 0;
    p3->n = 0;
    while(i < p1->n && j < p2->n){
        if(p1->t[i].exp > p2->t[j].exp){
            append(p3, p1->t[i].coef, p1->t[i].exp);
            i++;
        }
        else if(p1->t[i].exp < p2->t[j].exp){
            append(p3, p2->t[j].coef, p2->t[j].exp);
            j++;
        }
        else{
            int sum = p1->t[i].coef + p2->t[j].coef;
            if(sum != 0){
                append(p3, sum, p1->t[i].exp);
            }
            i++;
            j++;
        }
    }
    while(i < p1->n){
        append(p3, p1->t[i].coef, p1->t[i].exp);
        i++;
    }
    while(j < p2->n){
        append(p3, p2->t[j].coef, p2->t[j].exp);
        j++;
    }
}

```

```

void subtraction_polynomial(poly *p1, poly *p2, poly *p3){
    int i = 0, j = 0;
    p3->n = 0;
    while(i < p1->n && j < p2->n){
        if(p1->t[i].exp > p2->t[j].exp){
            append(p3, p1->t[i].coef, p1->t[i].exp);
            i++;
        }
        else if(p1->t[i].exp < p2->t[j].exp){
            append(p3, -p2->t[j].coef, p2->t[j].exp);
            j++;
        }
        else{
            int diff = p1->t[i].coef - p2->t[j].coef;
            if(diff != 0){
                append(p3, diff, p1->t[i].exp);
            }
            i++;
            j++;
        }
    }
    while(i < p1->n){
        append(p3, p1->t[i].coef, p1->t[i].exp);
        i++;
    }

    while(j < p2->n){
        append(p3, -p2->t[j].coef, p2->t[j].exp);
        j++;
    }
}

```

